

Math 242, S15
Quiz 11

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Use Laplace transform to solve the initial value problem:

$$x'' + 25x = 0; \quad x(0) = 4, \quad x'(0) = 5.$$

$$\text{Let } \mathcal{L}\{x(t)\} = X(s), \text{ then}$$

$$\mathcal{L}\{x''(t)\} = s^2 X(s) - 4s - 5 \quad \text{since } x(0) = 4, x'(0) = 5$$

$$\mathcal{L}\{x'' + 25x\} = \mathcal{L}\{0\}$$

$$[s^2 X(s) - 4s - 5] + 25X(s) = 0$$

$$(s^2 + 25)X(s) = 4s + 5$$

$$X(s) = \frac{4s + 5}{s^2 + 25}$$

$$\text{Thus } x(t) = \mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{4s + 5}{s^2 + 25}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{4s}{s^2 + 25}\right\} + \mathcal{L}^{-1}\left\{\frac{5}{s^2 + 25}\right\}$$

$$= 4 \cos 5t + \sin 5t$$

Problem 2. (4 points) Find the inverse Laplace transform of $\frac{1}{s(s^2 + 9)}$ using the formula $\mathcal{L}^{-1}\left(\frac{F(s)}{s}\right) = \int_0^t \mathcal{L}^{-1}\{F(s)\} d\tau$.

$$F(s) = \frac{1}{s^2 + 9} \Rightarrow \mathcal{L}^{-1}\{F(s)\} = \frac{1}{3} \sin 3t$$

$$\begin{aligned} \text{Thus } \mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + 9)}\right\} &= \int_0^t \frac{1}{3} \sin 3\tau d\tau \\ &= \left[-\frac{1}{9} \cos 3\tau\right]_{\tau=0}^{\tau=t} \\ &= \frac{1}{9} (1 - \cos 3t) \end{aligned}$$